high frequency products

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RF Amplifiers ACOM 600S, ACOM A1200S
Serial Port
Communication Protocol

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1. Structure of the messages between the amplifier and computer and protocol rules

Communication between the amplifier and computer is realized trough RS232 interface on the rear panel of the amplifier. Settings of RS232 interface are: baud rate 9600 8N1. Pause between messages is 10 ms. Max length of the messages is 72 bytes. When the computer receive message from the amplifier it have to return special confirmation message (message with address 0x86). If this message not received by the amplifier last message from the amplidier will be sent again. Each message initiated by the amplifier must be confirmed by the computer. Otherwise it will be sent again. If the message is not confirmed it will be discarded after the fifth attempt.

Structure of the communication protocol messages:

```
|Start TAG| Address | Length  |  - - - - D A T A - - - -  | CHKSUM  |
|0101 0101|xxxx xxxx|  N + 1  |xxxx xxxx|       |xxxx xxxx|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|       |---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | . . . | Byte N-1| Byte N  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – unique for each message
Byte 2 – Length of the message (all bytes including start byte ans check sum)
Byte 3..Byte N-1 – Data field
Byte N – Check Sum: CHKSUM

$Byte_{(N)} = 256 – \{SUM (Byte_{(0)} .. Byte_{(N-1)}) \& 0xFF\}$

If the message is received correctly $SUM (Byte_{(0)} .. Byte_{(N)}) = 0$.

2. Measurement units

Measurement units of parameters are noted in square brackets:

- [us] – microseconds
- [ms] – milliseconds
- [s] – seconds
- [mV] – millivolts
- [V] – volts
- [10 x V] – value is volts multiplied by 10. Example: 521 means U = 52.1 V
- [W] – watts
- [10 x W] – value is watts multiplied by 10. Example: 253 means U = 25.3 W
- [deg K] – temperature in degree Kelvins
- [deg C] – temperature in degree Celsius
- [deg F] – temperature in degree Fahreinheit
- [mA] – milliampers
- [A] – ampers
- [kHz] – kilohertz
- [100x] – value is the parameter multiplied by 100. Example: 217 means SWR = 2.17

3. Messages and commands from the Amplifier to the computer

3.1. Reply to the wrong received message:

Note: Generated after error in receiving!

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|0000 0001|0000 0100|1010 0110|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x01
Byte 2 – Length of the message – 4 Bytes
Byte 3 – CHKSUM of the message – 0xA6

3.2. Request for specific message:

```
|Start TAG| Address | Length  |  Data   | CHKSUM  |
|0101 0101|0000 0010|0000 0101|xxxx xxxx|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x02
Byte 2 – Length of the message – 5 bytes
Byte 3 – Number of desired message
Byte 4 – CHKSUM of the message

3.3. OK Message from the Amplifier:

```
|Start TAG| Address | Length  |Param (H)|Param (L)| CHKSUM  |
|0101 0101|0000 0011|0000 0110|pppp pppp|pppp pppp|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x03
Byte 2 – Length of the message – 6 bytes
Byte 3 – Return Parameter (High Byte)
Byte 4 – Return Parameter (Low Byte)
Byte 5 – CHKSUM of the message

3.4. ERR Message from the Amplifier:

```
|Start TAG| Address | Length  |Param (H)|Param (L)| CHKSUM  |
|0101 0101|0000 0100|0000 0110|pppp pppp|pppp pppp|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x04
Byte 2 – Length of the message – 6 bytes

Byte 3 – Return Parameter (High Byte)
Byte 4 – Return Parameter (Low Byte)
Byte 5 – CHKSUM of the message

3.5. System Configuration

3.5.1. PCB Versions of the main CPU module

Note: Generated once on request!

Start TAG	Address	Length	Ampl Type	FW Ver	FW SVer	CPU Ver
0101 0101	0001 0001	0001 1110	tttt tttt	vvvv vvvv	ssss ssss	xxxx xxxx
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

CPU SVer	IntA Ver	IntA SVer	IntB Ver	IntB SVer	Mod. Ver	Mod. SVer
xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13

BL Ver	BL SVer	A.SN [0]	A.SN [1]	A.SN [2]	A.SN [3]	A.SN [4]
xxxx xxxx	xxxx xxxx	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 14	Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20

A.SN [5]	A.SN [6]	A.SN [7]	A.SN [8]	A.SN [9]	A.SN [10]	A.SN [11]
ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 21	Byte 22	Byte 23	Byte 24	Byte 25	Byte 26	Byte 27

HF Count	CHKSUM
ffff ffff	cccc cccc
Byte 28	Byte 29

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x11
Byte 2 – Length of the message – 30 bytes
Byte 3 – Amplifier Type (1 - A600S, 2 – A1200S)
Byte 4 – Amplifier Firmware version
Byte 5 – Amplifier Frmware sub-version
Byte 6 – PCB CPU version
Byte 7 – PCB CPU sub-version
Byte 8 – PCB Interface_A version
Byte 9 – PCB Interface_A sub-version
Byte 10 – PCB Interface_B version
Byte 11 – PCB Interface_B sub-version
Byte 12 – PCB ATU Modem version (if the modem is mounted)
Byte 13 – PCB ATU Modem sub-version (if the modem is mounted)
Byte 14 –Bootloader version
Byte 15 –Bootloader sub-version
Byte 16..Byte 27 – Serial number of the amplifier
Byte 28 – Number of all Hard Fault records until now
Byte 29 – CHKSUM of the message

3.5.2. Settings of the Amplifier

Note: Generated once on request!

```
|Start TAG| Address | Length  |  CAT Settings 1   |  CAT Settings 2   |
|0101 0101|0001 0010|0000 1101|cccc cccc|cccc cccc|cccc cccc|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|  CAT Settings 3   |     User Flags    | Buz Vol | CHKSUM  |
|cccc cccc|cccc cccc|ffff ffff|ffff ffff|bbbb bbbb|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x12
Byte 2 – Length of the message – 13 bytes
Byte 3 – CAT Settings 1 (Low Byte)
    bit 4..bit 7 – CAT Baud Rate (valid only for CAT Interface with TTL Levels or RS232 Levels)
        0 – No CAT interface specified
        1 – 1200 bps
        2 – 4800 bps
        3 – 9600 bps
        4 – 19200 bps
        5 – 38400 bps
        6 – 57600 bps
    bit 0..bit 3 – Not Used
Byte 4 – CAT Settings 1 (High Byte)
    bit 4..bit 7 – CAT Interface Type (type of the CAT interface)
        0 – OFF
        1 – RS232 Levels
        2 – TTL Levels
        3 – BCD Code
        4 – RS232 Levels
    bit 0..bit 3 – CAT Command Set (Commands Set)
        0 – No CAT interface specified
        1 – Commands Set 1
        ...
        15 – Commands Set 15
Byte 5 – CAT Settings 2 (Low Byte)
Byte 6 – CAT Settings 2 (High Byte)
    CAT Byte Spacing [us] (0..65535)
Byte 7 – CAT Settings 3 (Low Byte)
Byte 8 – CAT Settings 3 (High Byte)
    CAT Polling Time [ms] (0..65535)
Byte 9 – User Flags (Low Byte):
    bit 7 – (0x07)
    bit 6 – (0x06) Operate Access — 0 – Disabled — 1 – Enabled
    bit 5 – (0x05) ON_BTN Pulses — 0 – Long ON / Short OFF — 1 – Short ON / Long OFF
    bit 4 – (0x04) ATAC Procedure: — 0 – Manual — 1 – Auto
    bit 3 – (0x03) CPU Buzzer: — 0 – Disabled — 1 – Enabled
    bit 2 – (0x02) Temperature Flag: — 0 – [º C] — 1 – [º F]
    bit 1 – (0x01) Auto Operate: — 0 – Disabled — 1 – Enabled
    bit 0 – (0x00) Extra Cooling: — 0 – Disabled — 1 – Enabled
Byte 10 – User Flags (High Byte):

bit 7 – (0x0F) Last Power OFF is because of HARD FAULT Error
bit 6 – (0x0E)
bit 5 – (0x0D)
bit 4 – (0x0C)
bit 3 – (0x0B)
bit 2 – (0x0A)
bit 1 – (0x09)
bit 0 – (0x08)

Byte 11 – Buzzer Volume (0 – 100%)
Byte 12 – CHKSUM of the message

3.5.3. Serial number of the amplifier:

Note: Generated once on request!

Start TAG	Address	Length	A.SN [0]	A.SN [1]	A.SN [2]	A.SN [3]	A.SN [4]
0101 0101	0001 0011	0001 0000	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

A.SN [5]	A.SN [6]	A.SN [7]	A.SN [8]	A.SN [9]	A.SN [10]	A.SN [11]	CHKSUM
ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	cccc cccc
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x13
Byte 2 – Length of the message – 16 bytes
Byte 3..Byte 14 – Serial number of the amplifier
Byte 15 – CHKSUM of the message

3.5.4. User Sign, saved in EEPROM memory of the main CPU of the amplifier

Note: Generated once on request!

Start TAG	Address	Length	H.Msg[0]	H.Msg[1]	H.Msg[2]	H.Msg[3]	H.Msg[4]
0101 0101	0001 0100	0001 0011	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

H.Msg[5]	H.Msg[6]	H.Msg[7]	H.Msg[8]	H.Msg[9]	H.Msg[10]	H.Msg[11]	H.Msg[12]
ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

H.Msg[13]	H.Msg[14]	CHKSUM
ssss ssss	ssss ssss	cccc cccc
Byte 16	Byte 17	Byte 18

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x14
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 17 – Hello Message – chars from 0 to 14
Byte 15 – CHKSUM of the message

```
|Start TAG| Address | Length  |H.Msg[15]|H.Msg[16]|H.Msg[17]|H.Msg[18]|H.Msg[19]|
|0101 0101|0001 0101|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[20]|H.Msg[21]|H.Msg[22]|H.Msg[23]|H.Msg[24]|H.Msg[25]|H.Msg[26]|H.Msg[27]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[28]|H.Msg[29]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x15
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 14 – Hello message – signs from 15 to 29
Byte 15 – CHKSUM of the message

3.5.5. ATU (Antenna Tuner) Parameters:

Note: Generated once on request!

```
|Start TAG| Address | Length  |ATU Type | HW Ver  | HW SVer | FW Ver  | FW SVer |
|0101 0101|0001 0110|0001 0011|tttt tttt|tttt tttt|ssss ssss|vvvv vvvv|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|ATUSN[0] |ATUSN[1] |ATUSN[2] |ATUSN[3] |ATUSN[4] |ATUSN[5] |ATUSN[6] |ATUSN[7] |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|ATUSN[8] |ATUSN[9] | CHKSUM  |
|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x16
Byte 2 – Length of the message – 19 bytes
Byte 3 – ATU/ASEL Type
    0x00 – No ATU/ASEL Assigned
    0x21 – ATU (Model 04AT) is assigned
    0x41 – ASEL (Model 10AS) is assigned
Byte 4 – ATU/ASEL Hardware version
Byte 5 – ATU/ASEL Hardware sub-version
Byte 6 – ATU/ASEL Firmware version
Byte 7 – ATU/ASEL Firmware sub-version
Byte 8, Byte 17 – Serial number of the ATU/ASEL
Byte 18 – CHKSUM of the message

3.5.6. ASEL (Antenna Selector) Parameters: (Obsolete! Message 0x16 is extended)

Note: Generated once on request!

```
|Start TAG| Address | Length  |ASEL Type| HW Ver  | HW SVer | FW Ver  | FW SVer |
|0101 0101|0001 0111|0001 0011|tttt tttt|tttt tttt|ssss ssss|vvvv vvvv|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|ASELSN[0]|ASELSN[1]|ASELSN[2]|ASELSN[3]|ASELSN[4]|ASELSN[5]|ASELSN[6]|ASELSN[7]|
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|ASELSN[8]|ASELSN[9]| CHKSUM  |
|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x17
Byte 2 – Length of the message – 19 bytes
Byte 3 – ASEL Type
0x00 – No ATU/ASEL Assigned0x21 – ATU (Model 04AT) is assigned
0x41 – ASEL (Model 10AS) is assigned
Byte 4 – ASEL hardware version
Byte 5 – ASEL hardware sub-version
Byte 6 – ASEL Firmware version
Byte 7 – ASEL Firmware sub-version
Byte 8, Byte 17 – Serial number of the ASEL
Byte 18 – CHKSUM of the message

3.5.7. Antenna Assignments saved in EEPROM memory in main CPU

Note: Generated once on request! (Similar to Message 0xA0)

```
|Start TAG| Address | Length  |Ant0 Type|Ant1 Type|Ant2 Type|Ant3 Type|Ant0 BndH|
|0101 0101|0001 1000|0001 0000|0000 0ttt|0000 0ttt|0000 0ttt|0000 0ttt|0000 0bbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|Ant0 BndL|Ant1 BndH|Ant1 BndL|Ant2 BndH|Ant2 BndL|Ant3 BndH|Ant3 BndL| CHKSUM  |
|bbbb bbbb|0000 0bbb|bbbb bbbb|0000 0bbb|bbbb bbbb|0000 0bbb|bbbb bbbb|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x18
Byte 2 – Length of the message – 16 bytes
Byte 3 ÷ Byte 6 – Antenna Type:
0 – Sharp, 1 – Regular, 2 – Wide, 3 – Fixed
Byte 7, Byte 8 – Antenna 0 Assignments (Band Low, Band High)
Byte 9, Byte 10 – Antenna 1 Assignments (Band Low, Band High)
Byte 11, Byte 12 – Antenna 2 Assignments (Band Low, Band High)
Byte 13, Byte 14 – Antenna 3 Assignments (Band Low, Band High)

bit 0 – 160 m / 1,8 MHz / 1800 kHz - 2000 kHz
bit 1 – 80 m / 3,5 MHz / 3500 kHz - 4000 kHz
bit 2 – 60 m / 5,0 MHz / 5060 kHz - 5450 kHz
bit 3 – 40 m / 7,0 MHz / 7000 kHz - 7300 kHz
bit 4 – 30 m / 10 MHz / 10100 kHz - 10150 kHz
bit 5 – 20 m / 14 MHz / 14000 kHz - 14350 kHz
bit 6 – 17 m / 18 MHz / 18068 kHz - 18168 kHz
bit 7 – 15 m / 21 MHz / 21000 kHz - 21450 kHz
bit 8 – 12 m / 24 MHz / 24890 kHz - 24990 kHz
bit 9 – 10 m / 28 MHz / 28000 kHz - 29700 kHz
bit 10 – 6 m / 50 MHz / 50000 kHz - 54000 kHz

Byte 15 – CHKSUM of the message

3.5.8. Fixed values of bias voltages of the transistors (Obsolete! Removed)

Note: Generated once on request!

```
|Start TAG| Address | Length  |  BIAS_1A_SET [mV] |  BIAS_1B_SET [mV] |
|0101 0101|0001 1001|0001 0100|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|  BIAS_1C_SET [mV] |  BIAS_1D_SET [mV] |  BIAS_2A_SET [mV] |  BIAS_2B_SET [mV] |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|  BIAS_2C_SET [mV] |  BIAS_2D_SET [mV] | CHKSUM  |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x19
Byte 2 – Length of the message – 20 bytes
Byte 3 – BIAS_1A_SET [mV] (Low Byte) (Not Used)
Byte 4 – BIAS_1A_SET [mV] (High Byte) (Not Used)
Byte 5 – BIAS_1B_SET [mV] (Low Byte) (Not Used)
Byte 6 – BIAS_1B_SET [mV] (High Byte) (Not Used)
Byte 7 – BIAS_1C_SET [mV] (Low Byte) (Not Used)
Byte 8 – BIAS_1C_SET [mV] (High Byte) (Not Used)
Byte 9 – BIAS_1D_SET [mV] (Low Byte) (Not Used)
Byte 10 – BIAS_1D_SET [mV] (High Byte) (Not Used)
Byte 11 – BIAS_2A_SET [mV] (Low Byte) (Not Used)
Byte 12 – BIAS_2A_SET [mV] (Hugh Byte) (Not Used)
Byte 13 – BIAS_2B_SET [mV] (Low Byte) (Not Used)
Byte 14 – BIAS_2B_SET [mV] (High Byte) (Not Used)
Byte 15 – BIAS_2C_SET [mV] (Low Byte) (Not Used)
Byte 16 – BIAS_2C_SET [mV] (High Byte) (Not Used)
Byte 17 – BIAS_2D_SET [mV] (Low Byte) (Not Used)
Byte 18 – BIAS_2D_SET [mV] (High Byte) (Not Used)
Byte 19 – CHKSUM of the message

3.6. ATU/ASEL Parameters

3.6.1. ATU Segments Tune Parameter:

Note: Generated once on request!

Start TAG	Address	Length	Band	Band_L_Freq [kHz]	Band_H_Freq [kHz]
0101 0101	0001 1010	0001 0110	bbbb bbbb	llll llll	llll llll	hhhh hhhh	hhhh hhhh
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

Seg_Cnt	Seg_Num	Ant_Num	Ant_Type	Segment_L_Freq	Segment_H_Freq
cccc cccc	nnnn nnnn	aaaa aaaa	tttt tttt	llll llll	llll llll	hhhh hhhh	hhhh hhhh
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

CT [%]	L [%]	CA [%]	Tuning Age [hours]	CHKSUM
tttt tttt	llll llll	aaaa aaaa	hhhh hhhh	hhhh hhhh	cccc cccc
Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x1A
Byte 2 – Length of the message – 22 bytes
Byte 3 – Band Number [1..11]
Byte 4, Byte 5 –Low Frequency of the Band [kHz]
Byte 6, Byte 7 – High Frequency of the Band [kHz]
Byte 8 – Number of Segments in the Band
Byte 9 – Number of Current Segment in the Band
Byte 10 – Antenna Number [1..10]
Byte 11 – Antenna Type
0 – Sharp, 1 – Regular, 2 – Wide, 3 – Fixed
Byte 12, Byte 13 –Low Frequency of the Segment [kHz]
Byte 14, Byte 15 – High Frequency of the Segment [kHz]
Byte 16 – Tuning CT [%]
Byte 17 – Tuning L [%]
Byte 18 – Tuning CA [%]
Byte 19, Byte 20 – Tuning Age [hours]
Byte 21 – CHKSUM of the message

3.6.2. Antenna Names Send to the Display:

Note: Generated once on request!

Start TAG	Address	Length	A1_Nm[0]	A1_Nm[1]	A1_Nm[2]	A1_Nm[3]	A1_Nm[4]
0101 0101	0001 1011	0011 0100	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

A1_Nm[5]	A1_Nm[6]	A1_Nm[7]	A1_Nm[8]	A1_Nm[9]	A1_Nm[10]	A1_Nm[11]	A2_Nm[0]
nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

```
|A2_Nm[1] |A2_Nm[2] |A2_Nm[3] |A2_Nm[4] |A2_Nm[5] |A2_Nm[6] |A2_Nm[7] |A2_Nm[8] |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 | Byte 21 | Byte 22 | Byte 23 |

|A2_Nm[9] |A2_Nm[10]|A2_Nm[11]|A3_Nm[0] |A3_Nm[1] |A3_Nm[2] |A3_Nm[3] |A3_Nm[4] |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 24 | Byte 25 | Byte 26 | Byte 27 | Byte 28 | Byte 29 | Byte 30 | Byte 31 |

|A3_Nm[5] |A3_Nm[6] |A3_Nm[7] |A3_Nm[8] |A3_Nm[9] |A3_Nm[10]|A3_Nm[11]|A4_Nm[0] |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 32 | Byte 33 | Byte 34 | Byte 35 | Byte 36 | Byte 37 | Byte 38 | Byte 39 |

|A4_Nm[1] |A4_Nm[2] |A4_Nm[3] |A4_Nm[4] |A4_Nm[5] |A4_Nm[6] |A4_Nm[7] |A4_Nm[8] |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 40 | Byte 41 | Byte 42 | Byte 43 | Byte 44 | Byte 45 | Byte 46 | Byte 47 |

|A4_Nm[9] |A4_Nm[10]|A4_Nm[11]| CHKSUM  |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|
| Byte 48 | Byte 49 | Byte 50 | Byte 51 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x1B
Byte 2 – Length of the message – 52 bytes
Byte 3 ÷ Byte 14 – Name of the Antenna 1
Byte 15 ÷ Byte 26 – Name of the Antenna 2
Byte 27 ÷ Byte 38 – Name of the Antenna 3
Byte 39 ÷ Byte 50 – Name of the Antenna 4
Byte 51 – CHKSUM of the message

3.7. Parameters of the amplifier:

3.7.1. Error Codes (ErrorCodes0 ... ErrorCodesA):

Note: Generated once when new alarm was appeared or at any time on request!

```
|Start TAG| Address | Length  |   Error Codes 0   |   Error Codes 1   |
|0101 0101|0010 0001|0001 1010|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|   Error Codes 2   |   Error Codes 3   |   Error Codes 4   |   Error Codes 5   |
|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|   Error Codes 6   |   Error Codes 7   |   Error Codes 8   |
|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |

|   Error Codes 9   |    Error Codes A  | CHKSUM  |
|eeee eeee|eeee eeee|eeee eeee|eeee eeee|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 21 | Byte 22 | Byte 23 | Byte 24 | Byte 25 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x21
Byte 2 – Length of the message – 26 bytes
Byte 3 – Error Codes 0 (Low Byte):

bit 7 - (0x07) EXCESSIVE DRIVE POWER	(SOFT FAULT)
bit 6 - (0x06) DRIVE POWER TOO HIGH	(WARNING)
bit 5 - (0x05) EXCESSIVE REFLECTED POWER	(SOFT FAULT)
bit 4 - (0x04) REFLECTED POWER WARNING	(WARNING)
bit 3 - (0x03) DRIVE POWER DETECTED AT WRONG TIME	(HARD FAULT)
bit 2 - (0x02) OUTPUT RELAY OPEN IT SHOULD BE CLOSED	(HARD FAULT)
bit 1 - (0x01) OUTPUT RELAY CLOSED IT SHOULD BE OPEN	(WARNING)
bit 0 - (0x00) HOT SWITCHING ATTEMPT	(WARNING)

Byte 4 – Error Codes 0 (High Byte):

bit 7 - (0x0F) REMOVE DRIVE POWER IMMEDIATELY	(WARNING)
bit 6 - (0x0E) STOP TRANSMISSION FIRST	(WARNING)
bit 5 - (0x0D) PA LOAD SWR TOO HIGH	(SOFT FAULT)
bit 4 - (0x0C) DETECTED RF POWER AT WRONG TIME	(WARNING)
bit 3 - (0x0B) OUTPUT DISBALANCE	(SOFT FAULT)
bit 2 - (0x0A) FREQUENCY VIOLATION	(SOFT FAULT)
bit 1 - (0x09) DRIVE FREQUENCY OUT OF RANGE	(SOFT FAULT)
bit 0 - (0x08) HOT SWITCHING ATTEMPT	(SOFT FAULT)

Byte 5 – Error Codes 1 (Low Byte):

bit 7 - (0x17) LPF FAN SPEED TOO LOW	(HARD FAULT)
bit 6 - (0x16) PAM2 FAN SPEED TOO LOW	(HARD FAULT)
bit 5 - (0x15) PAM1 FAN SPEED TOO LOW	(HARD FAULT)
bit 4 - (0x14)	
bit 3 - (0x13) 26V TOO HIGH	(HARD FAULT)
bit 2 - (0x12) 26V TOO LOW	(HARD FAULT)
bit 1 - (0x11) 5V TOO HIGH	(HARD FAULT)
bit 0 - (0x10) 5V TOO LOW	(HARD FAULT)

Byte 6 – Error Codes 1 (High Byte):

bit 7 - (0x1F) PAM2 EXCESSIVE TEMPERATURE	(SOFT FAULT)
bit 6 - (0x1E) PAM1 EXCESSIVE TEMPERATURE	(SOFT FAULT)
bit 5 - (0x1D) PAM2 TEMPERATURE TOO HIGH	(WARNING)
bit 4 - (0x1C) PAM1 TEMPERATURE TOO HIGH	(WARNING)
bit 3 - (0x1B) PAM2 DISSIPATION POWER WARNING	(WARNING)
bit 2 - (0x1A) PAM1 DISSIPATION POWER WARNING	(WARNING)
bit 1 - (0x19) PAM2 DISSIPATION POWER TOO HIGH	(SOFT FAULT)
bit 0 - (0x18) PAM1 DISSIPATION POWER TOO HIGH	(SOFT FAULT)

Byte 7 – Error Codes 2 (Low Byte):

bit 7 - (0x27) BIAS_1B VOLTAGE ERROR	(SOFT FAULT)
bit 6 - (0x26) BIAS_1A VOLTAGE ERROR	(SOFT FAULT)
bit 5 - (0x25) PAM1 EXCESSIVE CURRENT	(SOFT FAULT)
bit 4 - (0x24) PAM1 CURRENT WARNING	(WARNING)
bit 3 - (0x23) PAM1 IDLE CURRENT TOO LOW	(SOFT FAULT)
bit 2 - (0x22) PAM1 CURRENT = …mA SHOULD BE ZERO	(SOFT FAULT)
bit 1 - (0x21) PAM1 HV TOO HIGH	(HARD FAULT)
bit 0 - (0x20) PAM1 HV TOO LOW	(HARD FAULT)

Byte 8 – Error Codes 2 (High Byte):

bit 7 - (0x2F) PAM1 GAIN TOO HIGH	(SOFT FAULT)
bit 6 - (0x2E) PAM1 GAIN TOO LOW	(SOFT FAULT)
bit 5 - (0x2D) BIAS_1D = …V SHOULD BE ZERO	(HARD FAULT)
bit 4 - (0x2C) BIAS_1C = …V SHOULD BE ZERO	(HARD FAULT)
bit 3 - (0x2B) BIAS_1B = …V SHOULD BE ZERO	(HARD FAULT)

bit 2 - (0x2A) BIAS_1A = …V SHOULD BE ZERO (HARD FAULT)
bit 1 - (0x29) BIAS_1D VOLTAGE ERROR (SOFT FAULT)
bit 0 - (0x28) BIAS_1C VOLTAGE ERROR (SOFT FAULT)

Byte 9 – Error Codes 3 (Low Byte):

bit 7 - (0x37) BIAS_1D = …V SHOULD BE ZERO (HARD FAULT)
bit 6 - (0x36) BIAS_1C = …V SHOULD BE ZERO (HARD FAULT)
bit 5 - (0x35) BIAS_1B = …V SHOULD BE ZERO (HARD FAULT)
bit 4 - (0x34) BIAS_1A = …V SHOULD BE ZERO (HARD FAULT)
bit 3 - (0x33) PAM1 TEMPERATURE TOO HIGH (WARNING)
bit 2 - (0x32) PAM1 EXCESSIVE TEMPERATURE (HARD FAULT)
bit 1 - (0x31) PAM1 CURRENT = …mA SHOULD BE ZERO (HARD FAULT)
bit 0 - (0x30) PAM1 HV = …V SHOULD BE ZERO (HARD FAULT)

Byte 10 – Error Codes 3 (High Byte):

bit 7 - (0x3F)
bit 6 - (0x3E)
bit 5 - (0x3D)
bit 4 - (0x3C)
bit 3 - (0x3B)
bit 2 - (0x3A)
bit 1 - (0x39) PAM1 EXCESSIVE CURRENT. CHECK SWR OR REDUCE DRIVE POWER (SOFT FAULT)
bit 0 - (0x38) PSU1 EXCESSIVE TEMPERATURE (WARNING)

Byte 11 – Error Codes 4 (Low Byte):

bit 7 - (0x47) BIAS_2B VOLTAGE ERROR (SOFT FAULT)
bit 6 - (0x46) BIAS_2A VOLTAGE ERROR (SOFT FAULT)
bit 5 - (0x45) PAM2 EXCESSIVE CURRENT (SOFT FAULT)
bit 4 - (0x44) PAM2 CURRENT WARNING (WARNING)
bit 3 - (0x43) PAM2 IDLE CURRENT TOO LOW (SOFT FAULT)
bit 2 - (0x42) PAM2 CURRENT = …mA SHOULD BE ZERO (SOFT FAULT)
bit 1 - (0x41) PAM2 HV TOO HIGH (HARD FAULT)
bit 0 - (0x40) PAM2 HV TOO LOW (HARD FAULT)

Byte 12 – Error Codes 4 (High Byte):

bit 7 - (0x4F) PAM2 GAIN TOO HIGH (SOFT FAULT)
bit 6 - (0x4E) PAM2 GAIN TOO LOW (SOFT FAULT)
bit 5 - (0x4D) BIAS_2D = …V SHOULD BE ZERO (HARD FAULT)
bit 4 - (0x4C) BIAS_2C = …V SHOULD BE ZERO (HARD FAULT)
bit 3 - (0x4B) BIAS_2B = …V SHOULD BE ZERO (HARD FAULT)
bit 2 - (0x4A) BIAS_2A = …V SHOULD BE ZERO (HARD FAULT)
bit 1 - (0x49) BIAS_2D VOLTAGE ERROR (SOFT FAULT)
bit 0 - (0x48) BIAS_2C VOLTAGE ERROR (SOFT FAULT)

Byte 13 – Error Codes 5 (Low Byte):

bit 7 - (0x57) BIAS_2D = …V SHOULD BE ZERO (HARD FAULT)
bit 6 - (0x56) BIAS_2C = …V SHOULD BE ZERO (HARD FAULT)
bit 5 - (0x55) BIAS_2B = …V SHOULD BE ZERO (HARD FAULT)
bit 4 - (0x54) BIAS_2A = …V SHOULD BE ZERO (HARD FAULT)
bit 3 - (0x53) PAM2 TEMPERATURE TOO HIGH (WARNING)
bit 2 - (0x52) PAM2 EXCESSIVE TEMPERATURE (HARD FAULT)
bit 1 - (0x51) PAM2 CURRENT = …mA SHOULD BE ZERO (HARD FAULT)
bit 0 - (0x50) PAM2 HV = …V SHOULD BE ZERO (HARD FAULT)

Byte 14 – Error Codes 5 (High Byte):

bit 7 - (0x3F)
bit 6 - (0x5E)
bit 5 - (0x5D)
bit 4 - (0x5C)

bit 3 - (0x5B)
bit 2 - (0x5A)
bit 1 - (0x59) PAM2 EXCESSIVE CURRENT. CHECK SWR OR REDUCE DRIVE POWER (SOFT FAULT)
bit 0 - (0x58) PSU2 EXCESSIVE TEMPERTURE (WARNING)

Byte 15 – Error Codes 6 (Low Byte):
bit 7 - (0x67) ATU POWER SWITCH ALARM AT POWER ON (WARNING)
bit 6 - (0x66) ATU POWER SWITCH ALARM (WARNING)
bit 5 - (0x65) ATU MODEM EXCESSIVE TEMPERATURE (WARNING)
bit 4 - (0x64) DISPLAY UNIT COMMUNICATION ERROR (WARNING)
bit 3 - (0x63) PSU2 EXCESSIVE TEMPERATURE (SOFT FAULT)
bit 2 - (0x62) PSU1 EXCESSIVE TEMPERATURE (SOFT FAULT)
bit 1 - (0x61) PSU2/CONTROL MALFUNCTION (HARD FAULT)
bit 0 - (0x60) PSU1/CONTROL MALFUNCTION (HARD FAULT)

Byte 16 – Error Codes 6 (High Byte):
bit 7 - (0x6F) LOSS OF EEPROM DATA (SOFT FAULT)
bit 6 - (0x6E) LOSS OF EEPROM DATA (WARNING)
bit 5 - (0x6D) LOSS OF ETHERNET SETTINGS (WARNING)
bit 4 - (0x6C) LOSS OF AUDIO DATA (WARNING)
bit 3 - (0x6B)
bit 2 - (0x6A)
bit 1 - (0x69) AUDIO MEMORY NOT RESPONDING (WARNING)
bit 0 - (0x68) ETHERNET CONTROLLER NOT RESPONDING (WARNING)

Byte 17 – Error Codes 7 (Low Byte):
bit 7 - (0x77)
bit 6 - (0x76)
bit 5 - (0x75)
bit 4 - (0x74)
bit 3 - (0x73)
bit 2 - (0x72)
bit 1 - (0x71)
bit 0 - (0x70) CAT ERROR (WARNING)

Byte 18 – Error Codes 7 (High Byte):
bit 7 - (0x7F)
bit 6 - (0x7E)
bit 5 - (0x7D)
bit 4 - (0x7C)
bit 3 - (0x7B)
bit 2 - (0x7A)
bit 1 - (0x79)
bit 0 - (0x78)

Byte 19 – Error Codes 8 (Low Byte):
bit 7 - (0x87) NO ANTENNA SETTINGS PREPARED (WARNING)
bit 6 - (0x86) NO TUNUNG SETTINGS PREPARED (WARNING)
bit 5 - (0x85) AMP-ASEL COMMUNICATION ERROR (WARNING)
bit 4 - (0x84) ASEL-AMP COMMUNICATION ERROR (WARNING)
bit 3 - (0x83) ASEL NOT RESPONDING (WARNING)
bit 2 - (0x82) AMP-ATU COMMUNICATION ERROR (WARNING)
bit 1 - (0x81) ATU-AMP COMMUNICATION ERROR (WARNING)
bit 0 - (0x80) ATU NOT RESPONDING (WARNING)

Byte 20 – Error Codes 8 (High Byte):
bit 7 - (0x8F)
bit 6 - (0x8E)
bit 5 - (0x8D)

bit 4 - (0x8C)
bit 3 - (0x8B) ATU MEMORY FAIL (WARNING)
bit 2 - (0x8A) ATU TUNING CYCLE UNSUCCESSFUL (WARNING)
bit 1 - (0x89) ANTENNA CAN NOT BE CHANGED UNTIL RF POWER IS PRESENT (WARNING)
bit 0 - (0x88) ATU CAN NOT BE RE-TUNED UNTIL RF POWER IS PRESENT

Byte 21 – Error Codes 9 (Low Byte):
bit 7 - (0x87) BIAS_2D = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 6 - (0x86) BIAS_2C = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 5 - (0x85) BIAS_2B = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 4 - (0x84) BIAS_2A = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 3 - (0x83) BIAS_1D = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 2 - (0x82) BIAS_1C = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 1 - (0x81) BIAS_1B = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 0 - (0x80) BIAS_1A = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)

Byte 22 – Error Codes 9 (High Byte):
bit 7 - (0x8F) BIAS_2D = …V SHOULD BE BELOW H_TH V (WARNING)
bit 6 - (0x8E) BIAS_2C = …V SHOULD BE BELOW H_TH V (WARNING)
bit 5 - (0x8D) BIAS_2B = …V SHOULD BE BELOW H_TH V (WARNING)
bit 4 - (0x8C) BIAS_2A = …V SHOULD BE BELOW H_TH V (WARNING)
bit 3 - (0x8B) BIAS_1D = …V SHOULD BE BELOW H_TH V (WARNING)
bit 2 - (0x8A) BIAS_1C = …V SHOULD BE BELOW H_TH V (WARNING)
bit 1 - (0x89) BIAS_1B = …V SHOULD BE BELOW H_TH V (WARNING)
bit 0 - (0x88) BIAS_1A = …V SHOULD BE BELOW H_TH V (WARNING)

Byte 23 – Error Codes A (Low Byte):
bit 7 - (0xA7) ANTENNA CURRENT TOO HIGH: ATU/ASEL IS OFF DUE TO DAMAGE RISK (HARD FAULT)
bit 6 - (0xA6) ANTENNA CURRENT TOO HIGH: REDUCE POWER (SOFT FAULT)
bit 5 - (0xA5) ANTENNA VOLTAGE TOO HIGH: ATU/ASEL IS OFF DUE TO DAMAGE RISK (HARD FAULT)
bit 4 - (0xA4) ANTENNA VOLTAGE TOO HIGH: REDUCE POWER (SOFT FAULT)
bit 3 - (0xA3) ATU/ASEL INTERNAL +5V TOO HIGH (HARD FAULT)
bit 2 - (0xA2) ATU/ASEL INTERNAL +5V TOO LOW (HARD FAULT)
bit 1 - (0xA1) ATU/ASEL DC-INPUT VOLTAGE TOO LOW (HARD FAULT)
bit 0 - (0xA0) ATU/ASEL DC-INPUT VOLTAGE TOO HIGH (HARD FAULT)

Byte 24 – Error Codes A (High Byte):
bit 7 - (0xAF) ATU/ASEL TEMPERATURE TOO LOW: TURNED OFF DUE TO DAMAGE RISK (HARD FAULT)
bit 6 - (0xAE) ATU/ASEL TEMPERATURE TOO HIGH: TURNED OFF DUE TO DAMAGE RISK (HARD FAULT)
bit 5 - (0xAD) ANTENNA SWR TOO HIGH: ATU/ASEL DAMAGE RISK (WARNING)
bit 4 - (0xAC) ANTENNA SWR TOO HIGH: ATU/ASEL DAMAGE RISK (SOFT FAULT)
bit 3 - (0xAB) ATU/ASEL RF-INPUT POWER TOO HIGH: TURNED OFF DUE TO DAMAGE RISK (HARD FAULT)

bit 2 - (0xAA) ATU/ASEL RF-INPUT POWER TOO HIGH: DAMAGE RISK (SOFT FAULT)
bit 1 - (0xA9) ANTENNA REFLECTED POWER TOO HIGH: ATU/ASEL IS OFF DUE TO DAMAGE RISK (HARD FAULT)
bit 0 - (0xA8) ANTENNA REFLECTED POWER TOO HIGH: ATU/ASEL DAMAGE RISK (SOFT FAULT)

Byte 25 – CHKSUM of the message

3.7.2. Parameters of the amplifier (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |Amp_State| Amplifier Flags 1 | Amplifier Flags 2 |
|0101 0101|0010 0011|0001 0100|ssss ssss|ffff ffff|ffff ffff|ffff ffff|ffff ffff|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|  P_DC 1 [10 x W]  |  P_DC 2 [10 x W]  |System Clock-HW [s]|System Clock-LW [s]|
|PPPP PPPP|PPPP PPPP|pppp pppp|pppp pppp|hhhh hhhh|hhhh hhhh|llll llll|llll llll|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|   temp1  [deg K]  |   temp2  [deg K]  | CHKSUM  |
|tttt tttt|tttt tttt|tttt tttt|tttt tttt|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x23
Byte 2 – Length of the message – 21 bytes
Byte 3 – Amplifier's mode
- 0x1X – Amplifier Reset
- 0x2X – Init Mode
- 0x3X – Debug Mode
- 0x4X – Service Mode
- 0x5X – STB Mode
- 0x6X – OPR/RX Mode
- 0x7X – OPR/TX Mode
- 0x8X – ATAC (ATU Tune Antenna Change) Procedure
- 0x9X – Set System Parameters Menu
- 0xAX – Turn OFF Procedure
- 0xA2 – Extra Cooling State

Byte 4 – Amplifier Flags 1 (Low Byte)
- bit 7 – FLAG_BIAS_CTRL — Request for Bias Voltages ON/OFF Procedures
- bit 6 – FLAG_PwrOff_Request — Final Request for Power OFF
- bit 5 – FLAG_Soft_PwrOff — Request for Power OFF by Software
- bit 4 – FLAG_PWRON_RS — Amplifier is Turned ON by RS232
- bit 3 – FLAG_PWRON_RMT — Amplifier is Turned ON by Remote Voltage
- bit 2 – FLAG_PWRON_BTN — Amplifier is Turned ON by Panel Button
- bit 1 – FLAG_PSU2_TEMPHI — PSU2 Over-Heating
- bit 0 – FLAG_PSU1_TEMPHI — PSU1 Over-Heating

Byte 5 – Amplifier Flags 1 (High Byte)
- bit 15 – FLAG_Last_Com — Last Command Source (0 - Local / 1 - Remote)
- bit 14 – SPARE — Spare Flag

bit 13 – FLAG_FULLTUNE_REQ Request for ATU Full Tune Procedure
bit 12 – FLAG_HV_ON Status of High Voltage Control Signal
bit 11 – FLAG_HV_MON Enable/Disable High Voltage Monitoring
bit 10 – FLAG_HV_CTRL Request for High Voltage ON/OFF Procedures
bit 9 – FLAG_BIASEN Status of Bias Control Signal
bit 8 – FLAG_BIAS_MON Enable/Disable Bias Voltages Monitoring

Byte 6 – Amplifier Flags 2 (Low Byte)
bit 7 – FLAG_ORC_MON Enable/Disable ORC Monitoring
bit 6 – FLAG_OUTR Status of the Output Relay Control Signal
bit 5 – FLAG_OUTR_CTRL Request for Output Relay ON/OFF Procedures
bit 4 – FLAG_TX_ACCESS Disable/Enable "OPR/TX Mode" Access
bit 3 – FLAG_INR Status of the Input Relay Control Signal
bit 2 – FLAG_INR_CTRL Request for Input Relay ON/OFF Procedures
bit 1 – FLAG_KEYIN Detected Level of the KEY IN Signal
bit 0 – FLAG_KEYIN_MON Enable/Disable KEY IN Monitoring

Byte 7 – Amplifier Flags 2 (High Byte)
bit 15 – FLAG_WRN_DIS Disable Warning Messages
bit 14 – FLAG_SF_ERR_DIS Disable SOFT FAULT Errors
bit 13 – FLAG_HF_ERR_DIS Disable HARD FAULT Errors
bit 12 – FLAG_I_LKG_MON ON/OFF Alarms Check During Input Relay Switching
bit 11 – SPARE Spare Flag
bit 10 – SPARE Spare Flag
bit 9 – FLAG_LPF_TUN Indicate Whether LPFs are Tuned or Not
bit 8 – FLAG_ORC ORC Condition After ORC Detection Procedure

Byte 8 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (Low Byte)
Byte 9 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (High Byte)
Byte 10 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (Low Byte) (Not Used)
Byte 11 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (High Byte) (Not Used)
Byte 12 – System Clock – High Word [sec] (Low Byte)
Byte 13 – System Clock – High Word [sec] (High Byte)
Byte 14 – System Clock – Low Word [sec] (Low Byte)
Byte 15 – System Clock – Low Word [sec] (High Byte)
Byte 16 – Temperature of PAM1 [deg K] (Low Byte)
Byte 17 – Temperature of PAM1 [deg K] (High Byte)
Byte 18 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 19 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 20 – CHKSUM of the message

3.7.3. Input power, forward power and reflected power of the amplifier (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |Input Power [10xW] | Forward Power [W] |
|0101 0101|0010 0100|0001 0100|iiii iiii|iiii iiii|oooo oooo|oooo oooo|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|Reflected Power [W]|     SWR [100x]    |PAM1 Dis.Pwr.[10xW]|PAM2 Dis.Pwr.[10xW]|
|rrrr rrrr|rrrr rrrr|ssss ssss|ssss ssss|pppp pppp|pppp pppp|pppp pppp|pppp pppp|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |
```

```
|    bdata  [mV]    |  disbalance [mV]  |  CHKSUM |
|bbbb bbbb|bbbb bbbb|dddd dddd|dddd dddd|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x24
Byte 2 – Length of the message – 20 bytes
Byte 3 – Input Power [10 x W] (Low Byte)
Byte 4 – Input Power [10 x W] (High Byte)
Byte 5 – Forward Power [W] (Low Byte)
Byte 6 – Forward Power [W] (High Byte)
Byte 7 – Reflected Power [W] (Low Byte)
Byte 8 – Reflected Power [W] (High Byte)
Byte 9 – SWR [100x] (Low Byte)
Byte 10 – SWR [100x] (High Byte)
Byte 11 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte)
Byte 12 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte)
Byte 13 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte) (Not Used)
Byte 14 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte) (Not Used)
Byte 15 – bdata [mV] (Low Byte)
Byte 16 – bdata [mV] (High Byte)
Byte 17 – disbalance [mV] (Low Byte) (Not Used)
Byte 18 – disbalance [mV] (High Byte) (Not Used)
Byte 19 – CHKSUM of the message

3.7.4. Power supply voltages, currents and carrier frequency (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |      VCC5 [mV]    |     VCC26 [mV]    |
|0101 0101|0010 0101|0001 0101|vvvv vvvv|vvvv vvvv|vvvv vvvv|vvvv vvvv|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|    HV1 [10 x V]   |    HV2 [10 x V]   |     ID1 [mA]      |     ID2 [mA]      |
|HHHH HHHH|HHHH HHHH|hhhh hhhh|hhhh hhhh|iiii iiii|iiii iiii|iiii iiii|iiii iiii|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|  Frequency [kHz]  |ErrorCode|ErrPrm.LW|ErrPrm.HW| CHKSUM  |
|ffff ffff|ffff ffff|eeee eeee|pppp pppp|pppp pppp|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x25
Byte 2 – Length of the message – 21 bytes
Byte 3 – VCC5 [mV] (Low Byte)
Byte 4 – VCC5 [mV] (High Byte)
Byte 5 – VCC26 [10 x V] (Low Byte)
Byte 6 – VCC26 [10 x V] (High Byte)
Byte 7 – HV1 [10 x V] (Low Byte)
Byte 8 – HV1 [10 x V] (High Byte)

Byte 9 – HV2, only if PAM2 Exsist [10 x V] (Low Byte) (Not Used)
Byte 10 – HV2, only if PAM2 Exsist [10 x V] (High Byte) (Not Used)
Byte 11 – ID1 [mA] (Low Byte)
Byte 12 – ID1 [mA] (High Byte)
Byte 13 – ID2, only if PAM2 Exsist [mA] (Low Byte) (Not Used)
Byte 14 – ID2, only if PAM2 Exsist [mA] (High Byte) (Not Used)
Byte 15 – Carrier frequency [kHz] (Low Byte)
Byte 16 – Carrier frequency [kHz] (High Byte)
Byte 17 – Error code of the alarm condition displayed on the amplifier's screen
Byte 18 – Parameter of the alarm condition (Low Byte)
Byte 19 – Parameter of the alarm condition (High Byte)
Byte 20 – CHKSUM of the message

3.7.5. Bias voltages (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |    bias_1a [mV]   |    bias_1b [mV]   |
|0101 0101|0010 0110|0001 0000|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|    bias_1c [mV]   |    bias_1d [mV]   |    bias_2a [mV]   |    bias_2b [mV]   |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|    bias_2c [mV]   |    bias_2d [mV]   | CHKSUM  |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 |
```

Byte 0 – Start byte– 0x55
Byte 1 – Address – 0x26
Byte 2 – Length of the message – 20 bytes
Byte 3 – bias_1a [mV] (Low Byte) (A600S, A1200S)
Byte 4 – bias_1a [mV] (High Byte) (A600S, A1200S)
Byte 5 – bias_1b [mV] (Low Byte) (A600S, A1200S)
Byte 6 – bias_1b [mV] (High Byte) (A600S, A1200S)
Byte 7 – bias_1c [mV] (Low Byte) (Not Used)
Byte 8 – bias_1c [mV] (High Byte) (Not Used)
Byte 9 – bias_1d [mV] (Low Byte) (Not Used)
Byte 10 – bias_1d [mV] (High Byte) (Not Used)
Byte 11 – bias_2a [mV] (Low Byte) (Not Used)
Byte 12 – bias_2a [mV] (High Byte) (Not Used)
Byte 13 – bias_2b [mV] (Low Byte) (Not Used)
Byte 14 – bias_2b [mV] (High Byte) (Not Used)
Byte 15 – bias_2c [mV] (Low Byte) (Not Used)
Byte 16 – bias_2c [mV] (High Byte) (Not Used)
Byte 17 – bias_2d [mV] (Low Byte) (Not Used)
Byte 18 – bias_2d [mV] (High Byte) (Not Used)
Byte 19 – CHKSUM of the message

3.7.6. Antenna number, selected band and band borders

Note: Generated only when band or band segmend was changed!

```
|Start TAG| Address | Length  | Ant Num |Ant Type |  Band   |Segm Cnt |
|0101 0101|0010 0111|0001 0101|nnnn nnnn|tttt tttt|bbbb bbbb|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|Segm Num | Displ Band L_Freq | Displ Band H_Freq |Seg Stat |Seg Stat |
|nnnn nnnn|llll llll|llll llll|hhhh hhhh|hhhh hhhh|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 |

|Seg Stat |Seg Stat |Seg Stat |Seg Stat |Seg Stat |ATU_Label| CHKSUM  |
|hhhh hhhh|hhhh hhhh|ssss ssss|ssss ssss|ssss ssss|llll llll|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 14 | Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x27
Byte 2 – Length of the message – 21 bytes
Byte 3 – Number of the selected antenna [1..10]
Byte 4 – Type of the selected antenna [0x00 - Sharp, 0x01 - Regular, 0x02 – Wide, 0x03 - Fixed]
If MSB is set – selected antenna must be marked as „by default”
Byte 5 – Number of the selected band [1..11]
Byte 6 – Number of segments in selected band (from 1 to 80)
Byte 7 – Number of the current segment
Byte 8 – Low band (or Segment) border for visualization [kHz] (Low Byte)
Byte 9 – Low band (or Segment) border for visualization [kHz] (High Byte)
Byte 10 – High band (or Segment) border for visualization [kHz] (Low Byte)
Byte 11 – High band (or Segment) border for visualization [kHz] (High Byte)
Byte 12 – Byte 18 – band segments condition (0 – not tuned , 1 – tuned)
Byte 19 – ATU_Label

Number	Label	Internal Flag	Priority
0	No Label	-	-
1	ATU Initialization	Init_In_Progress	2
2	Recalling Settings	Mem_Tune_In_Progress	2
3	Matching Cycle…	Full_Tune_In_Progress	2
4	ATU MEM Transfer	MEM_RW_In_Progress	2
5	Completed	Procedure_Completed	3
6	ATU In Bypass	ATU_In_Bypass	4
7	ATU Matched	Excelent_Tuned	5
8	Coarse ATU Match	Good_Tuned	5
9	No Match Found	Bad_Tuned	5
10	Settings Outdated	Tuning_Outdated	6
11	Memory Settings	Tuned	7

Byte 20 – CHKSUM of the message

Note: Band borders are different, depends of whether amplifier is locked or not.
Note: For ATU_Label – Priority 1 is Highest.

3.7.7. Band segments condition (Obsolete! Icluded in Message 0x27)

Note: Generated once when band or band segmend was changed!

```
|Start TAG| Address | Length  |Seg Stat |Seg Stat |Seg Stat |Seg Stat |Seg Stat |
|0101 0101|0010 1000|0000 1110|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|Seg Stat |Seg Stat |Seg Stat |Seg Stat |Seg Stat | CHKSUM  |
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x28
Byte 2 – Length of the message – 14 bytes
Byte 3 – Byte 12 – band segments condition (0 – not tuned , 1 – tuned)
Byte 13 – CHKSUM of the message

3.7.8. Parameters for Menu: “ATU Service”

```
|Start TAG| Address | Length  |ATU Model| HW.Ver  |HW.SubVer| FW.Ver  |FW.SubVer|
|0101 0101|0000 1010|0001 1000|mmmm mmmm|hhhh hhhh|hhhh hhhh|ffff ffff|ffff ffff|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|W.Hours L|W.Hours H| ATU_CT %| ATU_L % | ATU_CA %|ATUSN[0] |ATUSN[1] |ATUSN[2] |
|hhhh hhhh|hhhh hhhh|tttt tttt|llll llll|aaaa aaaa|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|ATUSN[3] |ATUSN[4] |ATUSN[5] |ATUSN[6] |ATUSN[7] |ATUSN[8] |ATUSN[9] | CHKSUM  |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 | Byte 21 | Byte 22 | Byte 23 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x2A
Byte 2 – Length of the message – 24 bytes
Byte 3 – ATU Model:
    0x00 – No ATU/ASEL assigned
    0x21 – ATU (Model 04AT) is assigned
    0x41 – ASEL (Model 10AS) is assigned
Byte 4 – ATU Hardware version
Byte 5 – ATU Hardware sub-version
Byte 6 – ATU Firmware version
Byte 7 – ATU Firmware sub-version
Byte 8 – ATU Working Hours [hours] (Low Byte)
Byte 9 – ATU Working Hours [hours] (High Byte)
Byte 10 – ATU_SET_CT Parameter [0..100] %
Byte 11 – ATU_SET_L Parameter [0..100] %
Byte 12 – ATU_SET_CA Parameter [0..100] %
Byte 13, Byte 22 – Serial number of the ATU
Byte 23 – CHKSUM of the message

3.7.9. Parameters for Menu: "ATU Measure"

Note: Generated automaticly with 10 ms pause between messages instead 0x2F if "Sys_Flags.Displ_ATU_Measure_Menu" flag is set!

Start TAG	Address	Length	temp1 [deg K]	temp2 [deg K]		
0101 0101	0010 1110	0001 1111	tttt tttt	tttt tttt	tttt tttt	tttt tttt
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

Input Power [10x W]		Forward Power [W]		Reflected Power [W]		ATU PS [mV]	
iiii iiii	iiii iiii	ffff ffff	ffff ffff	rrrr rrrr	rrrr rrrr	vvvv vvvv	vvvv vvvv
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14

Antenna Voltage [V]		Ant Current [10xA]		Ant Fwd Pwr [W]		Ant Refl Pwr [W]	
uuuu uuuu	uuuu uuuu	iiii iiii	iiii iiii	ffff ffff	ffff ffff	rrrr rrrr	rrrr rrrr
Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22

ATU SWR [100 x SWR]		ATU Temp [deg K]		ATU Working Houts		Sys Stat.	CHKSUM
ssss ssss	ssss ssss	tttt tttt	tttt tttt	hhhh hhhh	hhhh hhhh	ssss ssss	cccc cccc
Byte 23	Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x2E
Byte 2 – Length of the message – 31 bytes
Byte 3 – Temperature of PAM1 [deg K] (Low Byte)
Byte 4 – Temperature of PAM1 [deg K] (High Byte)
Byte 5 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 6 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 7 – Input Power [10 x W] (Low Byte)
Byte 8 – Input Power [10 x W] (High Byte)
Byte 9 – Forward Power [W] (Low Byte)
Byte 10 – Forward Power [W] (High Byte)
Byte 11 – Reflected Power [W] (Low Byte)
Byte 12 – Reflected Power [W] (High Byte)
Byte 13 – ATU Power Supply [10 x V] (Low Byte)
Byte 14 – ATU Power Supply [10 x V] (High Byte)
Byte 15 – ATU Antenna Voltage [V] (Low Byte)
Byte 16 – ATU Antenna Voltage [V] (High Byte)
Byte 17 – ATU Antenna Current [10 x A] (Low Byte)
Byte 18 – ATU Antenna Current [10 x A] (High Byte)
Byte 19 – ATU Antenna Forward Power [W] (Low Byte)
Byte 20 – ATU Antenna Forward Power [W] (High Byte)
Byte 21 – ATU Antenna Reflected Power [W] (Low Byte)
Byte 22 – ATU Antenna Reflected Power [W] (High Byte)
Byte 23 – ATU SWR [100 x SWR] (Low Byte)
Byte 24 – ATU SWR [100 x SWR] (High Byte)
Byte 25 – ATU Temperature [deg K] (Low Byte)
Byte 26 – ATU Temperature [deg K] (High Byte)
Byte 27 – ATU Working Hours [hours] (Low Byte)
Byte 28 – ATU Working Hours [hours] (High Byte)
Byte 29 – System status flags
bit 7 –
bit 6 –

bit 5 –
bit 4 –
bit 3 – Default Antenna is Selected
bit 2 – Hide Segments Scale (Only if ATU is Assigned)
bit 1 – ATU Assignment notification (ATU_Flags_0.Flag.Assigned)
bit 0 – Error Code is for last Hard Fault Error (User_Flags, bit 15 is active)
Byte 30 – CHKSUM of the message

3.7.10. Parameter of the amplifier (united messages: 0x23, 0x24, 0x25 and 0x26)

Note: Generated automaticly with 10 ms pause between messages!

Start TAG	Address	Length	Amp_State	Amplifier Flags 1	Amplifier Flags 2		
0101 0101	0010 1111	0100 1000	ssss ssss	ffff ffff	ffff ffff	ffff ffff	ffff ffff
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

P_DC 1 [10 x W]		P_DC 2 [10 x W]		System Clock-HW [s]		System Clock-LW [s]	
PPPP PPPP	PPPP PPPP	pppp pppp	pppp pppp	hhhh hhhh	hhhh hhhh	llll llll	llll llll
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

temp1 [deg K]		temp2 [deg K]		Input Power [10x W]		Forward Power [W]	
tttt tttt	tttt tttt	tttt tttt	tttt tttt	iiii iiii	iiii iiii	oooo oooo	oooo oooo
Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22	Byte 23

Reflected Power [W]		SWR [100x]		PAM1 Dis.Pwr.[10xW]		PAM2 Dis.Pwr.[10xW]	
rrrr rrrr	rrrr rrrr	ssss ssss	ssss ssss	pppp pppp	pppp pppp	pppp pppp	pppp pppp
Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30	Byte 31

bdata [mV]		disbalance [mV]		VCC5 [mV]		VCC26 [mV]	
bbbb bbbb	bbbb bbbb	dddd dddd	dddd dddd	vvvv vvvv	vvvv vvvv	vvvv vvvv	vvvv vvvv
Byte 32	Byte 33	Byte 34	Byte 35	Byte 36	Byte 37	Byte 38	Byte 39

HV1 [10 x V]		HV2 [10 x V]		ID1 [mA]		ID2 [mA]	
HHHH HHHH	HHHH HHHH	hhhh hhhh	hhhh hhhh	iiii iiii	iiii iiii	iiii iiii	iiii iiii
Byte 40	Byte 41	Byte 42	Byte 43	Byte 44	Byte 45	Byte 46	Byte 47

Frequency [kHz]		bias_1a [mV]		bias_1b [mV]		bias_1c [mV]	
ffff ffff	ffff ffff	aaaa aaaa	aaaa aaaa	bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc
Byte 48	Byte 49	Byte 50	Byte 51	Byte 52	Byte 53	Byte 54	Byte 55

bias_1d [mV]		bias_2a [mV]		bias_2b [mV]		bias_2c [mV]	
dddd dddd	dddd dddd	aaaa aaaa	aaaa aaaa	bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc
Byte 56	Byte 57	Byte 58	Byte 59	Byte 60	Byte 61	Byte 62	Byte 63

bias_2d [mV]		ErrorCode	ErrPrm.LW	ErrPrm.HW	S.Flags	Sys Stat.	CHKSUM
dddd dddd	dddd dddd	pppp pppp	pppp pppp	pppp pppp	ffff ffff	ssss ssss	cccc cccc
Byte 64	Byte 65	Byte 66	Byte 67	Byte 68	Byte 69	Byte 70	Byte 71

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x2F

Byte 2 – Length of the message – 72 bytes
Byte 3 – Mode of the Amplifier
- 0x1X – Amplifier Reset
- 0x2X – Init Mode
- 0x3X – Debug Mode
- 0x4X – Service Mode
- 0x5X – STB Mode
- 0x6X – OPR/RX Mode
- 0x7X – OPR/TX Mode
- 0x8X – ATAC (ATU Tune Antenna Change) Procedure
- 0x9X – Set System Parameters Menu
- 0xAX – Turn OFF Procedure
- 0xA2 – Extra Cooling State

Byte 4 – Amplifier Flags 1 (Low Byte)

bit 7 – FLAG_BIAS_CTRL	Request for Bias Voltages ON/OFF Procedures
bit 6 – FLAG_PwrOff_Request	Final Request for Power OFF
bit 5 – FLAG_Soft_PwrOff	Request for Power OFF by Software
bit 4 – FLAG_PWRON_RS	Amplifier is Turned ON by RS232
bit 3 – FLAG_PWRON_RMT	Amplifier is Turned ON by Remote Voltage
bit 2 – FLAG_PWRON_BTN	Amplifier is Turned ON by Panel Button
bit 1 – FLAG_PSU2_TEMPHI	PSU2 Over-Heating
bit 0 – FLAG_PSU1_TEMPHI	PSU1 Over-Heating

Byte 5 – Amplifier Flags 1 (High Byte)

bit 15 – FLAG_Last_Com	Last Command Source (0 - Local / 1 - Remote)
bit 14 – SPARE	Spare Flag
bit 13 – FLAG_FULLTUNE_REQ	Request for ATU Full Tune Procedure
bit 12 – FLAG_HV_ON	Status of High Voltage Control Signal
bit 11 – FLAG_HV_MON	Enable/Disable High Voltage Monitoring
bit 10 – FLAG_HV_CTRL	Request for High Voltage ON/OFF Procedures
bit 9 – FLAG_BIASEN	Status of Bias Control Signal
bit 8 – FLAG_BIAS_MON	Enable/Disable Bias Voltages Monitoring

Byte 6 – Amplifier Flags 2 (Low Byte)

bit 7 – FLAG_ORC_MON	Enable/Disable ORC Monitoring
bit 6 – FLAG_OUTR	Status of the Output Relay Control Signal
bit 5 – FLAG_OUTR_CTRL	Request for Output Relay ON/OFF Procedures
bit 4 – FLAG_TX_ACCESS	Disable/Enable “OPR/TX Mode” Access
bit 3 – FLAG_INR	Status of the Input Relay Control Signal
bit 2 – FLAG_INR_CTRL	Request for Input Relay ON/OFF Procedures
bit 1 – FLAG_KEYIN	Detected Level of the KEY IN Signal
bit 0 – FLAG_KEYIN_MON	Enable/Disable KEY IN Monitoring

Byte 7 – Amplifier Flags 2 (High Byte)

bit 15 – FLAG_WRN_DIS	Disable Warning Messages
bit 14 – FLAG_SF_ERR_DIS	Disable SOFT FAULT Errors
bit 13 – FLAG_HF_ERR_DIS	Disable HARD FAULT Errors
bit 12 – FLAG_I_LKG_MON	ON/OFF Alarms Check During Input Relay Switching
bit 11 – SPARE	Spare Flag
bit 10 – SPARE	Spare Flag
bit 9 – FLAG_LPF_TUN	Indicate Whether LPFs are Tuned or Not
bit 8 – FLAG_ORC	ORC Condition After ORC Detection Procedure

Byte 8 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (Low Byte)
Byte 9 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (High Byte)
Byte 10 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (Low Byte) (Not Used)
Byte 11 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (High Byte) (Not Used)
Byte 12 – System Clock – High Word [sec] (Low Byte)

Byte 13 – System Clock – High Word [sec] (High Byte)
Byte 14 – System Clock – Low Word [sec] (Low Byte)
Byte 15 – System Clock – Low Word [sec] (High Byte)
Byte 16 – Temperature of PAM1 [deg K] (Low Byte)
Byte 17 – Temperature of PAM1 [deg K] (High Byte)
Byte 18 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 19 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 20 – Input Power [10 x W] (Low Byte)
Byte 21 – Input Power [10 x W] (High Byte)
Byte 22 – Forward Power [W] (Low Byte)
Byte 23 – Forward Power [W] (High Byte)
Byte 24 – Reflected Power [W] (Low Byte)
Byte 25 – Reflected Power [W] (High Byte)
Byte 26 – SWR [100x] (Low Byte)
Byte 27 – SWR [100x] (High Byte)
Byte 28 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte)
Byte 29 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte)
Byte 30 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte) (Not Used)
Byte 31 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte) (Not Used)
Byte 32 – bdata [mV] (Low Byte)
Byte 33 – bdata [mV] (High Byte)
Byte 34 – disbalance [mV] (Low Byte) (Not Used)
Byte 35 – disbalance [mV] (High Byte) (Not Used)
Byte 36 – VCC5 [mV] (Low Byte)
Byte 37 – VCC5 [mV] (High Byte)
Byte 38 – VCC26 [10 x V] (Low Byte)
Byte 39 – VCC26 [10 x V] (High Byte)
Byte 40 – HV1 [10 x V] (Low Byte)
Byte 41 – HV1 [10 x V] (High Byte)
Byte 42 – HV2, only if PAM2 Exsist [10 x V] (Low Byte) (Not Used)
Byte 43 – HV2, only if PAM2 Exsist [10 x V] (High Byte) (Not Used)
Byte 44 – ID1 [mA] (Low Byte)
Byte 45 – ID1 [mA] (High Byte)
Byte 46 – ID2, only if PAM2 Exsist [mA] (Low Byte) (Not Used)
Byte 47 – ID2, only if PAM2 Exsist [mA] (High Byte) (Not Used)
Byte 48 – Carrier frequency [kHz] (Low Byte)
Byte 49 – Carrier frequency [kHz] (High Byte)
Byte 50 – bias_1a [mV] (Low Byte) (A600S, A1200S)
Byte 51 – bias_1a [mV] (High Byte) (A600S, A1200S)
Byte 52 – bias_1b [mV] (Low Byte) (A600S, A1200S)
Byte 53 – bias_1b [mV] (High Byte) (A600S, A1200S)
Byte 54 – bias_1c [mV] (Low Byte) (Not Used)
Byte 55 – bias_1c [mV] (High Byte) (Not Used)
Byte 56 – bias_1d [mV] (Low Byte) (Not Used)
Byte 57 – bias_1d [mV] (High Byte) (Not Used)
Byte 58 – bias_2a [mV] (Low Byte) (Not Used)
Byte 59 – bias_2a [mV] (High Byte) (Not Used)
Byte 60 – bias_2b [mV] (Low Byte) (Not Used)
Byte 61 – bias_2b [mV] (High Byte) (Not Used)
Byte 62 – bias_2c [mV] (Low Byte) (Not Used)
Byte 63 – bias_2c [mV] (High Byte) (Not Used)
Byte 64 – bias_2d [mV] (Low Byte) (Not Used)
Byte 65 – bias_2d [mV] (High Byte) (Not Used)
Byte 66 – Error code of the alarm condition displayed on the amplifier’s screen

Byte 67 – Parameter of the alarm condition (Low Byte)
Byte 68 – Parameter of the alarm condition (High Byte)
Byte 69 – Status flags for Fan speed and active LPF channel
bit 4 – bit 7 FAN Speed [0..4]
bit 0 – bit 3 Active FLP channel (low pass filter) [0..10]
Byte 70 – System status flags
bit 7 –
bit 6 –
bit 5 –
bit 4 –
bit 3 – Default Antenna is Selected
bit 2 – Hide Segments Scale (Only if ATU is Assigned)
bit 1 – ATU Assignment notification (ATU_Flags_0.Flag.Assigned)
bit 0 – Error Code is for last Hard Fault Error (User_Flags, bit 15 is active)
Byte 71 – CHKSUM of the message

4. HARD FAULT event record

When Hard Fault Event occurred, list of parameters of the amplifier at this moment are saved in non volatile EEPROM memory in the main microcontroller of the amplifier. Information about Hard Fault event record are represents with four messages. Each of them is generated once on request!

4.1. Hard Fault Error Event - Record 1

Start TAG	Address	Length	HF Count	HF NUM	Error Codes 0	
0101 0101	0011 0001	0010 0110	cccc cccc	nnnn nnnn	eeee eeee	eeee eeee
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

Error Codes 1		Error Codes 2		Error Codes 3		Error Codes 4	
eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14

Error Codes 5		Error Codes 6		Error Codes 7		Error Codes 8	
eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee
Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22

Error Codes 9		X SPARE X		Amplifier State		Amp Jump State	
eeee eeee	eeee eeee	ssss ssss	ssss ssss	ssss ssss	ssss ssss	jjjj jjjj	jjjj jjjj
Byte 23	Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30

Clock HW		Clock LW		HF1 CHKSUM		CHKSUM	
hhhh hhhh	hhhh hhhh	llll llll	llll llll	hhhh hhhh	hhhh hhhh	cccc cccc	
Byte 31	Byte 32	Byte 33	Byte 34	Byte 35	Byte 36	Byte 37	

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x31
Byte 2 – Length of the message – 38 bytes

Byte 3 – Number of all Hard Fault Errors until now
Byte 4 – Number of Hard Fault Error in the EEPROM memory
Byte 5 – Error Codes 0 (Low Byte)
Byte 6 – Error Codes 0 (High Byte)
Byte 7 – Error Codes 1 (Low Byte)
Byte 8 – Error Codes 1 (High Byte)
Byte 9 – Error Codes 2 (Low Byte)
Byte 10 – Error Codes 2 (High Byte)
Byte 11 – Error Codes 3 (Low Byte)
Byte 12 – Error Codes 3 (High Byte)
Byte 13 – Error Codes 4 (Low Byte)
Byte 14 – Error Codes 4 (High Byte)
Byte 15 – Error Codes 5 (Low Byte)
Byte 16 – Error Codes 5 (High Byte)
Byte 17 – Error Codes 6 (Low Byte)
Byte 18 – Error Codes 6 (High Byte)
Byte 19 – Error Codes 7 (Low Byte)
Byte 20 – Error Codes 7 (High Byte)
Byte 21 – Error Codes 8 (Low Byte)
Byte 22 – Error Codes 8 (High Byte)
Byte 23 – Error Codes 9 (Low Byte)
Byte 24 – Error Codes 9 (High Byte)
Byte 25 – Spare
Byte 26 – Spare
Byte 27 – Amplifier mode (Low Byte)
Byte 28 – Amplifier mode (High Byte)
Byte 29 – Amplifier transition state (Low Byte)
Byte 30 – Amplifier transition state (High Byte)
Byte 31 – System counter at the Hard Fault event (High Word / Low Byte)
Byte 32 – System counter at the Hard Fault event (High Word / High Byte)
Byte 33 – System counter at the Hard Fault event (Low Word / Low Byte)
Byte 34 – System counter at the Hard Fault event (Low Word / High Byte)
Byte 35 –Check Sum of the Hard Fault Record 1 (Low Byte)
Byte 36 –Check Sum of the Hard Fault Record 1 (High Byte)
Byte 37 – CHKSUM of the message

4.2. Hard Fault Error Event – Record 2

```
|Start TAG| Address | Length  |HF Count | HF NUM  |   IN FREQ [kHz]   |
|0101 0101|0011 0010|0010 0110|cccc cccc|nnnn nnnn|iiii iiii|iiii iiii|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|   Fwd Power [W]   |   Refl Power [W]  | In Power [10 x W] |  Disbalance [mV]  |
|ffff ffff|ffff ffff|rrrr rrrr|rrrr rrrr|iiii iiii|iiii iiii|dddd dddd|dddd dddd|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|  VCC26 [10 x V]   |     VCC5 [mV]     |   HV1 [10 x V]    |   HV2 [10 x V]    |
|VVVV VVVV|VVVV VVVV|vvvv vvvv|vvvv vvvv|hhhh hhhh|hhhh hhhh|hhhh hhhh|hhhh hhhh|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 | Byte 21 | Byte 22 |
```

```
|     ID1 [mA]      |     ID2 [mA]      |   TEMP1 [deg K]   |   TEMP2 [deg K]   |
|IIII IIII|IIII IIII|iiii iiii|iiii iiii|TTTT TTTT|TTTT TTTT|tttt tttt|tttt tttt|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 23 | Byte 24 | Byte 25 | Byte 26 | Byte 27 | Byte 28 | Byte 29 | Byte 30 |

|  P_DC1 [10 x W]   |  P_DC2 [10 x W]   |    HF2 CHKSUM     | CHKSUM  |
|DDDD DDDD|DDDD DDDD|dddd dddd|dddd dddd|hhhh hhhh|hhhh hhhh|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 31 | Byte 32 | Byte 33 | Byte 34 | Byte 35 | Byte 36 | Byte 37 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x32
Byte 2 – Length of the message – 38 bytes
Byte 3 – Number of all Hard Fault errors until now
Byte 4 – Number of Hard Fault error in the EEPROM memory
Byte 5 – Carrier frequency [kHz] (Low Byte)
Byte 6 – Carrier frequency [kHz] (High Byte)
Byte 7 – Forward Power [W] (Low Byte)
Byte 8 – Forward Power [W] (High Byte)
Byte 9 – Reflected Power [W] (Low Byte)
Byte 10 – Reflected Power [W] (High Byte)
Byte 11 – Input Power [10 x W] (Low Byte)
Byte 12 – Input Power [10 x W] (High Byte)
Byte 13 – Disbalance [mV] (Low Byte) (Not Used)
Byte 14 – Disbalance [mV] (High Byte) (Not Used)
Byte 15 – VCC26 [10 x V] (Low Byte)
Byte 16 – VCC26 [10 x V] (High Byte)
Byte 17 – VCC5 [mV] (Low Byte)
Byte 18 – VCC5 [mV] (High Byte)
Byte 19 – HV1 [10 x V] (Low Byte)
Byte 20 – HV1 [10 x V] (High Byte)
Byte 21 – HV2, only if PAM2 Exsist [10 x V] (Low Byte) (Not Used)
Byte 22 – HV2, only if PAM2 Exsist [10 x V] (High Byte) (Not Used)
Byte 23 – ID1 [mA] (Low Byte)
Byte 24 – ID1 [mA] (High Byte)
Byte 25 – ID2, only if PAM2 Exsist [mA] (Low Byte) (Not Used)
Byte 26 – ID2, only if PAM2 Exsist [mA] (High Byte) (Not Used)
Byte 27 – Temperature of PAM1 [deg K] (Low Byte)
Byte 28 – Temperature of PAM1 [deg K] (High Byte)
Byte 29 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 30 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 31 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (Low Byte)
Byte 32 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (High Byte)
Byte 33 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (Low Byte) (Not Used)
Byte 34 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (High Byte) (Not Used)
Byte 35 – Check Sum of the Hard Fault Record 2 (Low Byte)
Byte 36 – Check Sum of the Hard Fault Record 2 (High Byte)
Byte 37 – CHKSUM of the message

4.3. Hard Fault Error Event – Record 3

```
|Start TAG| Address | Length  |HF Count | HF NUM  |   BIAS_1A [mV]    |
|0101 0101|0011 0011|0010 0110|cccc cccc|nnnn nnnn|aaaa aaaa|aaaa aaaa|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |
```

BIAS_1B [mV]	BIAS_1C [mV]	BIAS_1D [mV]	BIAS_2A [mV]
bbbb bbbb | bbbb bbbb	cccc cccc | cccc cccc	dddd dddd | dddd dddd	aaaa aaaa | aaaa aaaa
Byte 7 | Byte 8	Byte 9 | Byte 10	Byte 11 | Byte 12	Byte 13 | Byte 14

BIAS_2B [mV]	BIAS_2C [mV]	BIAS_2D [mV]	P_DIS1 [10 x W]
bbbb bbbb | bbbb bbbb	cccc cccc | cccc cccc	dddd dddd | dddd dddd	dddd dddd | dddd dddd
Byte 15 | Byte 16	Byte 17 | Byte 18	Byte 19 | Byte 20	Byte 21 | Byte 22

P_DIS2 [10 x W]	SWR [100x]	bdata [mV]	CAT Settings 1
ssss ssss | ssss ssss	bbbb bbbb | bbbb bbbb	cccc cccc | cccc cccc	cccc cccc | cccc cccc
Byte 23 | Byte 24	Byte 25 | Byte 26	Byte 27 | Byte 28	Byte 29 | Byte 30

CAT Settings 2	CAT Settings 3	HF3 CHKSUM	CHKSUM
cccc cccc | cccc cccc	cccc cccc | cccc cccc	hhhh hhhh | hhhh hhhh	cccc cccc
Byte 31 | Byte 32	Byte 33 | Byte 34	Byte 35 | Byte 36	Byte 37

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x33
Byte 2 – Length of the message – 38 bytes
Byte 3 – Number of all Hard Fault errors until now
Byte 4 – Number of Hard Fault error in the EEPROM memory
Byte 5 – bias_1a [mV] (Low Byte) (A600S, A1200S)
Byte 6 – bias_1a [mV] (High Byte) (A600S, A1200S)
Byte 7 – bias_1b [mV] (Low Byte) (A600S, A1200S)
Byte 8 – bias_1b [mV] (High Byte) (A600S, A1200S)
Byte 9 – bias_1c [mV] (Low Byte) (Not Used)
Byte 10 – bias_1c [mV] (High Byte) (Not Used)
Byte 11– bias_1d [mV] (Low Byte) (Not Used)
Byte 12 – bias_1d [mV] (High Byte) (Not Used)
Byte 13 – bias_2a [mV] (Low Byte) (Not Used)
Byte 14 – bias_2a [mV] (High Byte) (Not Used)
Byte 15 – bias_2b [mV] (Low Byte) (Not Used)
Byte 16 – bias_2b [mV] (High Byte) (Not Used)
Byte 17 – bias_2c [mV] (Low Byte) (Not Used)
Byte 18 – bias_2c [mV] (High Byte) (Not Used)
Byte 19 – bias_2d [mV] (Low Byte) (Not Used)
Byte 20 – bias_2d [mV] (High Byte) (Not Used)
Byte 21 – P_DIS1 [10 x W] (Low Byte) (A600S, A1200S)
Byte 22 – P_DIS1 [10 x W] (High Byte) (A600S, A1200S)
Byte 23 – P_DIS2 [10 x W] (Low Byte) (Not Used)
Byte 24 – P_DIS2 [10 x W] (High Byte) (Not Used)
Byte 25 – SWR [100x] (Low Byte)
Byte 26 – SWR [100x] (High Byte)
Byte 27 – Analog Band Data [mV] (Low Byte)
Byte 28 – Analog Band Data [mV] (High Byte)
Byte 29 – CAT Settings 1 (Low Byte) –
bit 4..bit 7 – CAT baud rate (only for CAT interface with TTL Levels or RS232 Levels)
1 – 1200 bps
2 – 4800 bps
3 – 9600 bps
4 – 19200 bps
5 – 38400 bps

6 – 57600 bps

bit0..bit3 – Not Used

Byte 30 – CAT Settings 1 (High Byte)

bit 4..bit 7 – CAT Interface Type

0 – OFF

1 – RS232 Levels

2 – TTL Levels

3 – BCD Code

4 – RS232 Levels

bit0..bit3 – CAT Command Set

0 – CAT interface not specified

1 – Set 1

...

15 – Set 15

Byte 31 – CAT Settings 2 (Low Byte)

Byte 32 – CAT Settings 2 (High Byte) – Byte Spacing [us] (0..65535)

Byte 33 – CAT Settings 3 (Low Byte)

Byte 34 – CAT Settings 3 (High Byte) – Polling Time [ms] (0..65535)

Byte 35 – Check Sum of the Hard Fault Record 3 (Low Byte)

Byte 36 – Check Sum of the Hard Fault Record 3 (High Byte)

Byte 37 – CHKSUM of the message

4.4. Hard Fault Error Event – Record 4

Start TAG	Address	Length	HF Count	HF NUM	BIAS_1A_SET [mV]	
0101 0101	0011 0100	0010 0110	cccc cccc	nnnn nnnn	aaaa aaaa	aaaa aaaa
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

BIAS_1B_SET [mV]		BIAS_1C_SET [mV]		BIAS_1D_SET [mV]		BIAS_2A_SET [mV]	
bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc	dddd dddd	dddd dddd	aaaa aaaa	aaaa aaaa
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14

BIAS_2B_SET [mV]		BIAS_2C_SET [mV]		BIAS_2D_SET [mV]		Error Source	
bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc	dddd dddd	dddd dddd	eeee eeee	eeee eeee
Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22

X	SPARE	X	Aset	Atyp Anum	User Flags		LPF State	
xxxx xxxx	xxxx xxxx	ssss ssss	tttt nnnn	ffff ffff	ffff ffff	llll llll	llll llll	
Byte 23	Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30	

Amplifier Flags 1		Amplifier Flags 2		HF4 CHKSUM		CHKSUM	
ffff ffff	ffff ffff	ffff ffff	ffff ffff	hhhh hhhh	hhhh hhhh	cccc cccc	
Byte 31	Byte 32	Byte 33	Byte 34	Byte 35	Byte 36	Byte 37	

Byte 0 – Start byte of the message – 0x55

Byte 1 – Address of the message – 0x34

Byte 2 – Length of the message – 38 bytes

Byte 3 – Number of all Hard Fault errors until now

Byte 4 – Number of Hard Fault error in the EEPROM memory

Byte 5 – BIAS_1A_SET [mV] (Low Byte) (Not Used)

Byte 6 – BIAS_1A_SET [mV] (High Byte) (Not Used)

Byte 7 – BIAS_1B_SET [mV] (Low Byte) (Not Used)
Byte 8 – BIAS_1B_SET [mV] (High Byte) (Not Used)
Byte 9 – BIAS_1C_SET [mV] (Low Byte) (Not Used)
Byte 10 – BIAS_1C_SET [mV] (High Byte) (Not Used)
Byte 11 – BIAS_1D_SET [mV] (Low Byte) (Not Used)
Byte 12 – BIAS_1D_SET [mV] (High Byte) (Not Used)
Byte 13 – BIAS_2A_SET [mV] (Low Byte) (Not Used)
Byte 14 – BIAS_2A_SET [mV] (Hugh Byte) (Not Used)
Byte 15 – BIAS_2B_SET [mV] (Low Byte) (Not Used)
Byte 16 – BIAS_2B_SET [mV] (High Byte) (Not Used)
Byte 17 – BIAS_2C_SET [mV] (Low Byte) (Not Used)
Byte 18 – BIAS_2C_SET [mV] (High Byte) (Not Used)
Byte 19 – BIAS_2D_SET [mV] (Low Byte) (Not Used)
Byte 20 – BIAS_2D_SET [mV] (High Byte) (Not Used)
Byte 21 – Error Source (Low Byte)
Byte 22 – Error Source (High Byte)
Byte 23 – SPARE
Byte 24 – SPARE
Byte 25 – ATU Settings (Low Byte). Valid only when ATU is Associated
Byte 26 – ATU Settings (High Byte). Valid only when ATU is Associated
Byte 27 – User Flags (Low Byte):

bit 7 – (0x07)		
bit 6 – (0x06) Operate Access	0 – Disabled	1 – Enabled
bit 5 – (0x05) ON_BTN Pulses	0 – Long ON / Short OFF	1 – Short ON / Long OFF
bit 4 – (0x04) ATAC Procedure:	0 – Manual	1 – Auto
bit 3 – (0x03) CPU Buzzer:	0 – Disabled	1 – Enabled
bit 2 – (0x02) Temperature Flag:	0 – [º C]	1 – [º F]
bit 1 – (0x01) Auto Operate:	0 – Disabled	1 – Enabled
bit 0 – (0x00) Extra Cooling:	0 – Disabled	1 – Enabled

Byte 28 – User Flags (High Byte):
bit 7 – (0x0F) Last Power OFF is because of HARD FAULT Error
bit 6 – (0x0E)
bit 5 – (0x0D)
bit 4 – (0x0C)
bit 3 – (0x0B)
bit 2 – (0x0A)
bit 1 – (0x09)
bit 0 – (0x08)
Byte 29 – State of the Low Pass Filters (Low Byte)
Byte 30 – State of the Low Pass Filters (High Byte)
Byte 31 – Amplifier Flags 1
Byte 32 – Amplifier Flags 1
Byte 33 – Amplifier Flags 2
Byte 34 – Amplifier Flags 2
Byte 35 – Check Sum of the Hard Fault Record 4 (Low Byte)
Byte 36 – Check Sum of the Hard Fault Record 4 (High Byte)
Byte 37 – CHKSUM of the message

5. Messages and commands from the computer to the main CPU of the amplifier

5.1. Commands for amplifier management:

Start TAG	Address	Length	Amp CMD	Param. HW	Param. LW	Buttons	CHKSUM
0101 0101	1000 0001	0000 1000	cccc cccc	pppp pppp	pppp pppp	bbbb bbbb	cccc cccc
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Bute 7

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x81
Byte 2 – Length of the message – 8 bytes
Byte 3 – Command to the amplifier
- 0x00 – Empty command
- 0x01 – Request for specific message
  - Byte 4 – Number of desired message
  - Byte 5 – Number of desired message
- 0x02 – Request for amplifier mode change
  - Byte 4 – No matter
  - Byte 5 – Number of desired amplifier mode:
    - 0x01 – Amplifier restart
    - 0x02 – Amplifier restart
    - 0x03 – Debug Mode
    - 0x04 – Service Mode
    - 0x05 – STB Mode
    - 0x06 – OPR/RX Mode
    - 0x07 – OPR/TX Mode
    - 0x08 – Not Used
    - 0x09 – Menu “Set System Parameters”
    - 0x0A – Turn OFF the Amplifier
    - 0x40 – Prohibit OPR/TX mode
    - 0x80 – Allow OPR/TX mode
- 0x03 – Request for specific Hard Fault Event Record
  - Byte 4 – No matter
  - Byte 5 – Number of HARD FAULT event record
- 0x04 – Update user settings: User Flags
  - Byte 4 – User_Flags (HW)
  - Byte 5 – User_Flags (LW)
    - bit 15 - Last Power OFF is because of HARD FAULT Error (system flag)
    - bit 14 -
    - bit 13 -
    - bit 12 -
    - bit 11 -
    - bit 10 -
    - bit 9 -
    - bit 8 -
    - bit 7 -

bit 6 - Operate Access	0 – Disabled	1 – Enabled
bit 5 - ON_BTN Pulses	0 - Long ON / Short OFF	1 - Short ON / Long OFF
bit 4 - ATAC Procedure	0 - Manual	1 - Auto
bit 3 - CPU Buzzer	0 - Disabled	1 - Enabled
bit 2 - Temperaure Flag	0 - [deg C]	1 - [deg F]
bit 1 - Auto Operate	0 - Disabled	1 - Enabled
bit 0 - Extra Cooling	0 - Disabled	1 - Enabled

0x05 –CAT interface setting: CAT Settings 1:
- Byte 4 – CAT Settings 1 (HW)
- Byte 5 – CAT Settings 1 (LW)
  - bits [15..12] - CAT Interface Type:
    - 0 - None
    - 1 - RS232
    - 2 - TTL
    - 3 - BCD
    - 4 - Analog
  - bits [11..8] – CAT Command Set:
    - 0 - None
    - 1 - ICOM
    - 2 - YEASU FT-450, ...
    - 3 - YEASU FT-817, FT-897, ...
    - 4 - YEASU FT-1000MP, ...
    - 5 - ELECRAFT K3, KENWOOD TS-480 ...
    - 6 - ...
  - bits [7..4] - CAT_Baud_Rate:
    - 0 - None
    - 1- 1200
    - 2 - 4800
    - 3 - 9600
    - 4 - 19200
    - 5 - 38400
    - 6 - 57600

0x06 –CAT interface setting: CAT Settings 2: Byte Spacing [us]
- Byte 4 – CAT Settings 2 (HW)
- Byte 5 – CAT Settings 2 (LW)

0x07 –CAT interface setting: CAT Settings 3: Polling Time [ms]
- Byte 4 – CAT Settings 3 (HW)
- Byte 5 – CAT Settings 3 (LW)

0x08 – Clear SOFT FAULT errors and flag for last Power OFF is because of HARD FAULT Error (User Flags, bit 15)
- Byte 4 – No matter
- Byte 5 – No matter

0x09 – Command for manual changing of antenna and band
- Byte 4 – No Matter
- Byte 5 – Number of band:
  - 0x80 – Command “Next Band”
  - 0x40 – Command „Previous Band”
  - 0x20 – Command “Next Segment”
  - 0x10 – Command “Previous Segment”
  - 0x30 – Command “Next Antenna”

0x0A – Command for Buzzer volume and Buzzer sound signal generation
- Byte 4 – Sound signal choice
  - 0 – No signal
  - 1 – Signal BUZZER_PULSE (Single Pulse)
  - 2 – Signal BUZZER_WRN (Warning signal)
  - 3 – Signal BUZZER_PWROFF (Amplifier Power OFF signal)
  - 4 – Signal BUZZER_SF (Signal for SOFT FAULT error)
  - 5 – Signal BUZZER_HF (Signal for HARD FAULT error)
  - 6 – Signal BUZZER_PWRON (Amplifier Power ON signal)
  - ASCII – BUZZER_LETTER (Morze code coresponding to the ASCII)
- Byte 5 –Buzzer volume (0 – 100 %)

0x0B – Command for entire LOG file to be sent trough RS232 interface of the amplifier

Byte 4 – No matter

Byte 5 – No matter

0x0C – Commands when amplifier is in “Service Mode”

Byte 4 – Select module test

Byte 5 – Module test control

Byte 4		**Byte 5**		**Note**
0x00	Invalid value	0x00	Invalid Value	
0x01	Idle Current	0x00 0x01 0xFF	OFF ON Deactivate all tests	
0x02	High Voltage	0x00 0x01 0xFF	OFF ON Deactivate all tests	
0x03	Relay Voltage	0xFF	Deactivate all tests	Not Used
0x04	Band Select Relays Tests	0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0A 0x80 0x40 0xFF	OFF 160m 80m 40m 30m 20m 17m 15m 12m 10m 6m Command: Next Command: Previous Deactivate all tests	 United with 15m United with 17m United with 10m United with 12m
0x05	Input Relay Test	0x00 0x01	OFF ON	CPU Board returns the result
0x06	Output Relay Test	0x00 0x01	OFF ON	CPU Board returns the result
0x07	FAN Speed	0x00 0x01 0x02 0x03 0x80 0x40 0xFF	OFF Speed 1 Speed 2 Speed 3 Command: Next Command: Previous Deactivate all tests	If the temperature is high, command for lower fan speeds might be ignored.
0xFF	Deactivate all tests	X		

0x0D – Commands for default factory settings restore

Byte 4 – No matter

Byte 5 – Select the parameters to be restored

1 – Reset User Preferences to Factory Defaults

2 – Reset Working Hours Counter

3 – Erase Errors Log

4 – Reset All to Factory Defaults

5 – Restore „Antenna Assignments” to Factory Settings

0x0E – ATU Management

Byte 4 – ATU Command's Parammeter
Byte 5 – ATU Command
0x01 – ATU Full Tune Procedure. (Byte 4 value no matter)
0x02 – Bypass Command (Number of Antenna)
0x2E – Activate Cyclic Message 0x2E to the Display
0x2F – Activate Cyclic Message 0x2F to the Display
0xFF – Assign/Unassign ATU Status (Byte 4: 0x00 – ATU is unassigned, 0x21 – ATU is assigned)

0x0F – Request for specific ATU Tuning Record Settings
Byte 4
Bits7..4 – Number of Look-up Antenna [1..10]
Bits 3..0 – Number of the Look-up Band [1..11]
Byte 5 – Number of the Look-up ATU Tuning Record

0x10 – ATU Service Commands
Byte 4 – Command type:
0x01 – Erase Current Tuning (Byte 5 no matters)
0x02 – Erase Specific Antenna Tunings (Ant No. is in Byte 5)
0x04 – Erase All Tunings
0x08 – Erase Whole ATU Memory
0x10 – ATU "Memory Dump in a File" Command
0x20 – ATU "Memory Load from File" Command
0xFF – Exit from ATU Service menu
Byte 5 – Antenna Number in case of Erase Specific Antenna Tunings [1..10]

0x11 – Set "Tuning Outdated: time [h]
Byte4 – Outdated Time (High Byte)
Byte5 – Outdated Time (Low Byte)

0x12 – Delete Tuning of Specific Segment
Byte 4
Bits7..4 – Number of Look-up Antenna [1..10]
Bits 3..0 – Number of the Look-up Band [1..11]
Byte 5 – Number of the Look-up ATU Tuning Record

Byte 4 – The parameter depends of the command - Byte 3. (HIGH WORD)
Byte 5 – The parameter depends of the command - Byte 3. (LOW WORD)
Byte 6 – Amplifier's keyboard state - **Obsolete!**
bit 7 –
bit 6 –
bit 5 – Button 6 state
bit 4 – Button 5 state
bit 3 – Button 4 state
bit 2 – Button 3 state
bit 1 – Button 2 state
bit 0 – Button 1 state
Byte 7 – CHKSUM of the message

5.2. Serial number of the amplifier

```
|Start TAG| Address | Length  |A.SN [0] |A.SN [1] |A.SN [2] |A.SN [3] |A.SN [4] |
|0101 0101|1000 0010|0001 0000|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|A.SN [5] |A.SN [6] |A.SN [7] |A.SN [8] |A.SN [9] |A.SN [10]|A.SN [11]| CHKSUM  |
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x82
Byte 2 – Length of the message – 16 bytes
Byte 3..Byte 14 – Serial number of the amplifier
Byte 15 – CHKSUM of the message

5.3. Hello messages of the amplifier.

```
|Start TAG| Address | Length  |H.Msg[0] |H.Msg[1] |H.Msg[2] |H.Msg[3] |H.Msg[4] |
|0101 0101|1000 0011|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[5] |H.Msg[6] |H.Msg[7] |H.Msg[8] |H.Msg[9] |H.Msg[10]|H.Msg[11]|H.Msg[12]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[13]|H.Msg[14]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x83
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 17 – Hello message – chars from 0 to 14
Byte 18 – CHKSUM of the message

```
|Start TAG| Address | Length  |H.Msg[15]|H.Msg[16]|H.Msg[17]|H.Msg[18]|H.Msg[19]|
|0101 0101|1000 0100|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[20]|H.Msg[21]|H.Msg[22]|H.Msg[23]|H.Msg[24]|H.Msg[25]|H.Msg[26]|H.Msg[27]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[28]|H.Msg[29]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x84
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 17 – Hello message – chars from 15 to 29
Byte 18 – CHKSUM of the message

5.4. Antenna Names Received from the Display

```
|Start TAG| Address | Length  |A1_Nm[0] |A1_Nm[1] |A1_Nm[2] |A1_Nm[3] |A1_Nm[4] |
|0101 0101|1000 0101|0011 0100|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |
```

A1_Nm[5]	A1_Nm[6]	A1_Nm[7]	A1_Nm[8]	A1_Nm[9]	A1_Nm[10]	A1_Nm[11]	A2_Nm[0]
nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

A2_Nm[1]	A2_Nm[2]	A2_Nm[3]	A2_Nm[4]	A2_Nm[5]	A2_Nm[6]	A2_Nm[7]	A2_Nm[8]
nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22	Byte 23

A2_Nm[9]	A2_Nm[10]	A2_Nm[11]	A3_Nm[0]	A3_Nm[1]	A3_Nm[2]	A3_Nm[3]	A3_Nm[4]
nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30	Byte 31

A3_Nm[5]	A3_Nm[6]	A3_Nm[7]	A3_Nm[8]	A3_Nm[9]	A3_Nm[10]	A3_Nm[11]	A4_Nm[0]
nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 32	Byte 33	Byte 34	Byte 35	Byte 36	Byte 37	Byte 38	Byte 39

A4_Nm[1]	A4_Nm[2]	A4_Nm[3]	A4_Nm[4]	A4_Nm[5]	A4_Nm[6]	A4_Nm[7]	A4_Nm[8]
nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn	nnnn nnnn
Byte 40	Byte 41	Byte 42	Byte 43	Byte 44	Byte 45	Byte 46	Byte 47

A4_Nm[9]	A4_Nm[10]	A4_Nm[11]	CHKSUM
nnnn nnnn	nnnn nnnn	nnnn nnnn	cccc cccc
Byte 48	Byte 49	Byte 50	Byte 51

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x85
Byte 2 – Length of the message – 52 bytes
Byte 3 ÷ Byte 14 – Name of the Antenna 1
Byte 15 ÷ Byte 26 – Name of the Antenna 2
Byte 27 ÷ Byte 38 – Name of the Antenna 3
Byte 39 ÷ Byte 50 – Name of the Antenna 4
Byte 51 – CHKSUM of the message

5.5. Acknowledge after message receiving by main CPU module of the amplifier:

Generates automaticly after message receiving by main CPU module of the amplifier. If main CPU module don’t receive this message as reply last message will be sent again.

Start TAG	Address	Length	Msg Num	CHKSUM
0101 0101	1000 0110	0000 0101	nnnn nnnn	cccc cccc
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x86
Byte 2 – Length of the message – 5 bytes
Byte 3 – Number of received message
Byte 4 – CHKSUM of the message

5.6. Command for HEX messages format to ASCII messages format switching:

Note: Obsolete! (Removed)

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0000|0000 0100|0001 0111|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x90
Byte 2 – Length of the message – 4 bytes
Byte 3 – CHKSUM of the message – 0x17

Note: From ASCII message format to HEX messages format is switches with sign '~'

5.7. Disable Automatic telemetry data through RS232 interface of the amplifier:

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0001|0000 0100|0001 0110|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x91
Byte 2 – Length ofteh amplifier – 4 bytes
Byte 3 – CHKSUM of the message – 0x16

5.8. Enable automatic telemetry data through RS232 interface of the amplifier:

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0010|0000 0100|0001 0101|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x92
Byte 2 – Length of the message – 4 bytes
Byte 3 – CHKSUM of the message – 0x15

5.9. Bootloader activation:

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0011|0000 0100|0001 0100|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x93
Byte 2 – Length of the message – 4 bytes
Byte 3 – CHKSUM of the message – 0x14

5.10. Antenna Assinments:

(Similar to Message 0x18)

Start TAG	Address	Length	Ant0 Type	Ant1 Type	Ant2 Type	Ant3 Type	Ant0 BndH
0101 0101	1010 0000	0001 0000	0000 0ttt	0000 0ttt	0000 0ttt	0000 0ttt	0000 0bbb
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

Ant0 BndL	Ant1 BndH	Ant1 BndL	Ant2 BndH	Ant2 BndL	Ant3 BndH	Ant3 BndL	CHKSUM
bbbb bbbb	0000 0bbb	bbbb bbbb	0000 0bbb	bbbb bbbb	0000 0bbb	bbbb bbbb	cccc cccc
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0xA0
Byte 2 – Length of the message – 16 bytes
Byte 3 ÷ Byte 6 – Antenna Type:
0 – Sharp, 1 – Regular, 2 – Wide, 3 – Fixed
Byte 7, Byte 8 – Antenna 0 Assignments (Band Low, Band High)
Byte 9, Byte 10 – Antenna 1 Assignments (Band Low, Band High)
Byte 11, Byte 12 – Antenna 2 Assignments (Band Low, Band High)
Byte 13, Byte 14 – Antenna 3 Assignments (Band Low, Band High)
bit 0 – 160 m / 1,8 MHz / 1800 kHz - 2000 kHz
bit 1 – 80 m / 3,5 MHz / 3500 kHz - 4000 kHz
bit 2 – 60 m / 5,0 MHz / 5060 kHz - 5450 kHz
bit 3 – 40 m / 7,0 MHz / 7000 kHz - 7300 kHz
bit 4 – 30 m / 10 MHz / 10100 kHz - 10150 kHz
bit 5 – 20 m / 14 MHz / 14000 kHz - 14350 kHz
bit 6 – 17 m / 18 MHz / 18068 kHz - 18168 kHz
bit 7 – 15 m / 21 MHz / 21000 kHz - 21450 kHz
bit 8 – 12 m / 24 MHz / 24890 kHz - 24990 kHz
bit 9 – 10 m / 28 MHz / 28000 kHz - 29700 kHz
bit 10 – 6 m / 50 MHz / 50000 kHz - 54000 kHz
Byte 15 – CHKSUM of the message